



PROVINCIAL EXAMINATION

JUNE 2024

GRADE 6

MARKING GUIDELINES

NATURAL SCIENCES AND TECHNOLOGY
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6 pages

SECTION A		MARKS
QUESTION 1: Multiple Choice		5
QUESTION 2: Matching Items		5
QUESTION 3: Fill in the missing words		5
QUESTION 4: Labelling diagram		5
QUESTION 5: Labelling/Naming		10
TOTAL SECTION A:		30
SECTION B		MARKS
QUESTION 6: Case study, Compare, Analyse		5
QUESTION 7: Arrange		6
QUESTION 8: Explain, Compare, Illustrate		10
TOTAL SECTION B:		21
SECTION C		MARKS
QUESTION 9: Predict and Explain		2
QUESTION 10: Evaluate and Explain		3
QUESTION 11: Case Study/Investigation		4
TOTAL SECTION C:		9
TOTAL:		60

SECTION A	
QUESTION 1	MARKS
1.1 A ✓	1
1.2 C ✓	1
1.3 B ✓	1
1.4 C ✓	1
1.5 D ✓	1
	[5]
QUESTION 2	
2.1 E ✓	1
2.2 D ✓	1
2.3 A ✓	1
2.4 B ✓	1
2.5 C ✓	1
	[5]
QUESTION 3	
3.1 Contaminated ✓	1
3.2 Settling ✓	1
3.3 Purification ✓	1
3.4 Sludge ✓	1
3.5 Parasites ✓	1
	[5]

QUESTION 4		MARKS
4.1	Sieving ✓	1
4.2	Hand sorting ✓	1
4.3	Settling ✓	1
4.4	Filtering ✓	1
4.5	Decanting ✓	1
		[5]
QUESTION 5		
5.1	water ✓	1
5.2	water vapour ✓	1
5.3	ice ✓	1
5.4	A gas ✓ B liquid ✓ C Solid ✓	3
5.5	5.5.1 condensation ✓ 5.5.2 freezing/solidifying ✓ 5.5.3 melting ✓ 5.5.4 evaporation ✓	
		[10]
TOTAL SECTION A:		30

SECTION B

QUESTION 6

6.1 6.1.1 Piped water is water that has been cleaned and purified and channelled through pipes. ✓

6.1.2 Stream water is water that comes directly from a stream or river source and is not purified or cleaned. ✓

2

6.2 People use the bushes near the stream as a toilet hence the urine flows into the river/stream thus polluting it. ✓

1

6.3 6.3.1 Hand pumps can be used to pump up clean water used to clean one's hands or for personal use. ✓

6.3.2 Toilets are necessary as they channels the dirty water separately into another area thus not contaminating other sources of clean water. ✓

2

[5]

QUESTION 7

Step 1: F ✓

Step 2: C ✓

Step 3: E ✓

Step 4: A ✓

Step 5: B ✓

Step 6: D ✓

6

[6]

QUESTION 8

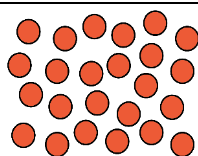
8.1 *Saturated solution*: A solution with a solute that is unable to dissolve anymore, leaving the undissolved substance at the bottom. ✓
Unsaturated solution: A solution which contains less than the maximum amount of solute that is capable of being dissolved. ✓

2

8.2 *Solids*: Particles that are very close together/there are small spaces between the particles/the particles are packed in a regular pattern/ the particles vibrate in a fixed position. ✓✓ (Any two)
Gases: Particles are very far apart from each other/there are big spaces between the particles/ particles move freely in all directions and move very fast. ✓✓ (Any two)

4

8.3 8.3.1



2

8.3.2 Liquid particles have medium spaces between the particles. Particles are fluid ✓✓ (Any two)

2

[10]

TOTAL SECTION B: 21

SECTION C	
QUESTION 9	
9.1 Yes, ✓ the substances do not settle. ✓ OR The Substance cannot be seen. ✓ OR The solution looks glassy ✓ (Any one)	2
	[2]
QUESTION 10	
Beaker B ✓ Hotter temperatures have more energy ✓ so they collide with the salt particles quicker, ✓ thus dissolving them much quicker.	3
	[3]
QUESTION 11	
11.1 Humans are the main cause of water pollution, ✓ they contaminate water by littering and industrial waste. ✓	1
11.2 Sieving: The holes in the sieve will clean water by allowing the water to pass through, while keeping the large substances behind. Filtering: Can separate bigger substances like plants, gravel and sand by putting a cloth over a container and pouring dirty water through the cloth. Settling and decanting: Allow solid substances to drop to the bottom of the container and carefully tilt the container at an angle. Boiling: Boiling is used to kill germs and parasites. Add chemicals: Adding just the correct amount of chlorine to dirty water to purify it. (Any one) ✓	1
11.3 Humans and animals need clean water to drink to stay healthy. Plants need water for photosynthesis. Some plants and animals need clean water in which to live and grow. Humans need clean water for cooking, cleaning and recreational activities like swimming. ✓✓ (Any suitable answer)	2
	[4]
TOTAL SECTION:	9
TOTAL:	60